MTH 132(C and K) Dr. James Ashe

Midterm Tuesday 4^{th} March, 2014 **Name:**

You must show work to receive credit. Do not use a calculator/computer for problems with a ★. Each problem is of equal value.

Find the slope of the line passing through the given pair of points .

1. $(1, 2)$ and $(3, -4)$

Use the given conditions to write an equation for the line in slope-intercept form, if possible.

2. Slope$= -2$, passing through $(13, 4)$.

3. Passing through $(-3, 0)$ and $(-6, 4)$.

Find the equation of the least squares line.

4. The paired data below consists of the number or Florida pleasure boats (in tens of thousands) and the number of yearly watercraft-related manatee deaths from 1995-2000.

Hours (x)	68	67	70	73	81	84
Score (y)	53	35	49	60	67	78

Solve the system of two equations in two unknowns.

5.
$$\begin{aligned} x + y &= 5 \\ x - y &= 10 \end{aligned}$$

6.
$$\begin{aligned} x + y &= 2 \\ 2x + 2y &= 5 \end{aligned}$$

Solve the problem.

7. There are 1000 people at a basketball game. The admission prices was \$20 for non-students and \$10 for students. The admission receipts were \$17,500. How many students and non-students attended the game?

8. Two pounds of rice and three pounds of potatoes cost a total of \$8, and three pounds of rice and one pound of potatoes cost a total of \$5. What is the price per pound for rice and the price per pound of potatoes?

9. A car manufacturer has decided to come out with a new type of car. The fixed cost for the production of this new car is \$250,000 and the variable costs are \$15,000 per car. The company expects to sell the cars for \$20,000 each. Formulate a function $C(x)$ for the total cost of producing x new cars and a function $R(x)$ for the total revenue generated by the sales of x cars.

10. After 5 years as an actuary, Sue's salary was \$100,000. After 10 years her salary had risen to \$200,000. Let y represent her salary after x years on the job. Assuming that a change in her salary over time can be approximated by a straight line, give an equation for this line in the form $y = mx + b$.

11. A package delivery company charges \$5.00 per package to deliver. The fixed cost to run the delivery truck is \$450 per day. If the company charges \$7.00 per package, how many packages must be delivered to make a profit of \$500?

12. A package delivery company charges \$5 per package to deliver. The fixed cost to run the delivery truck is \$450 per day. If the company charges \$7 per package, how many packages must be delivered to break even?

13. Let the supply and demand functions for gasoline be given by

$$p = S(q) = \frac{4}{5}q \text{ and } p = D(q) = 300 - \frac{2}{5}q$$

where p is the price in dollars and q is the number of 42-gallon barrels. Find the equilibrium quantity and the equilibrium price.

The sizes of two matrices A and B are given. Find the sizes of the product AB and the product BA, whenever these products exist.

14. A is 2×5, and B is 5×3.

a. AB:

b. BA:

Find the matrix if possible.

15. $\begin{bmatrix} 3 & 0 \\ -2 & 1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \end{bmatrix}$

16. $\begin{bmatrix} 3 & 0 \\ -2 & 1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$

17. $\begin{bmatrix} a & b & 1 \\ 0 & c & 0 \\ 0 & 1 & d \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ where a, b, c, d are real numbers.

Solve the problem.

18. Let $A = \begin{bmatrix} -1 & 3 \\ 2 & 0 \end{bmatrix}$ Find $-2A$.

19. Let $A = \begin{bmatrix} -1 & 3 \\ 2 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 2 \\ 5 & -1 \end{bmatrix}$. Find $A + B$.

20. Let $A = \begin{bmatrix} -1 & 3 \\ 2 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 2 \\ 5 & -1 \end{bmatrix}$. Find $2A + B$.

Bonus

★ Find the inverse (A^{-1}) of $A = \begin{bmatrix} 1 & 0 \\ 2 & 2 \end{bmatrix}$ without using a calculator/computer.